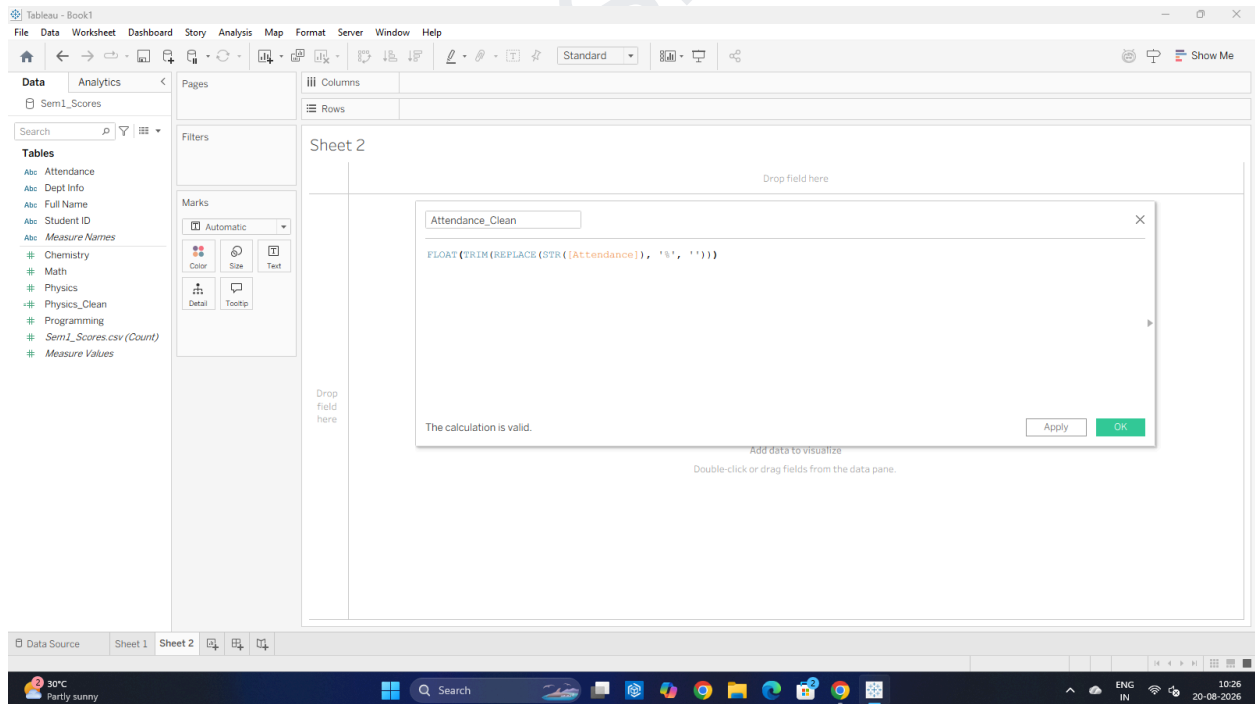
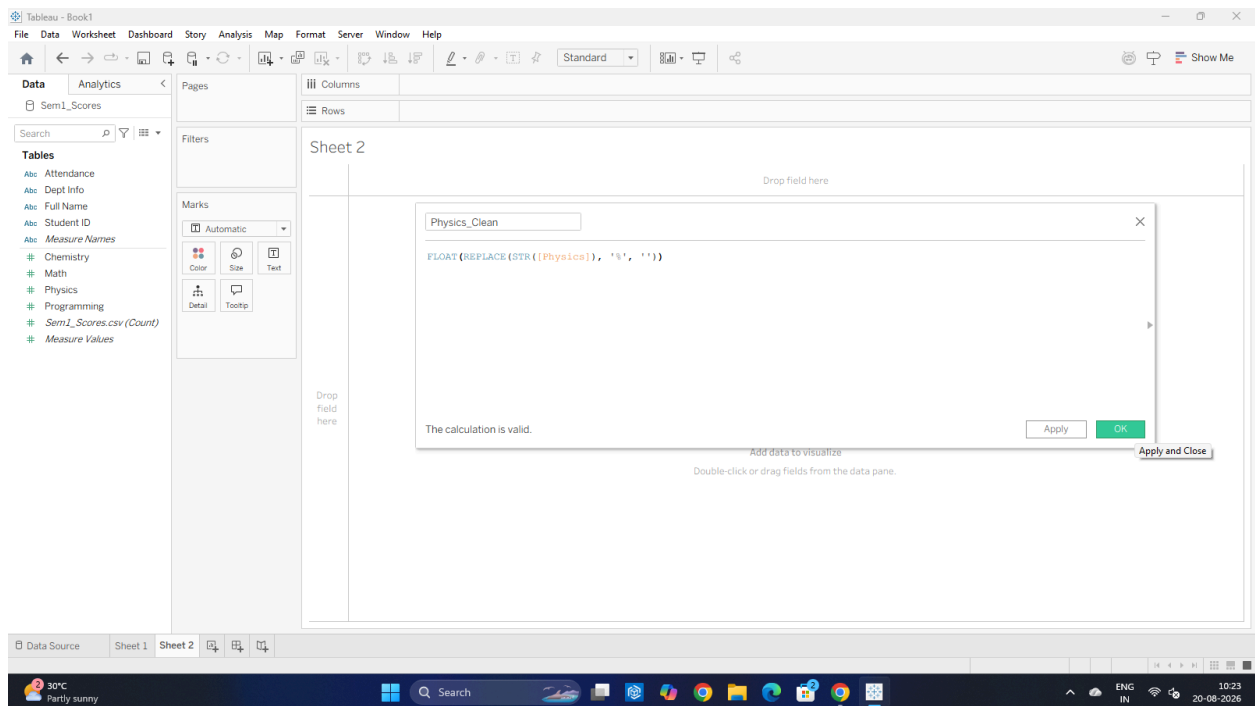


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Aim: Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data



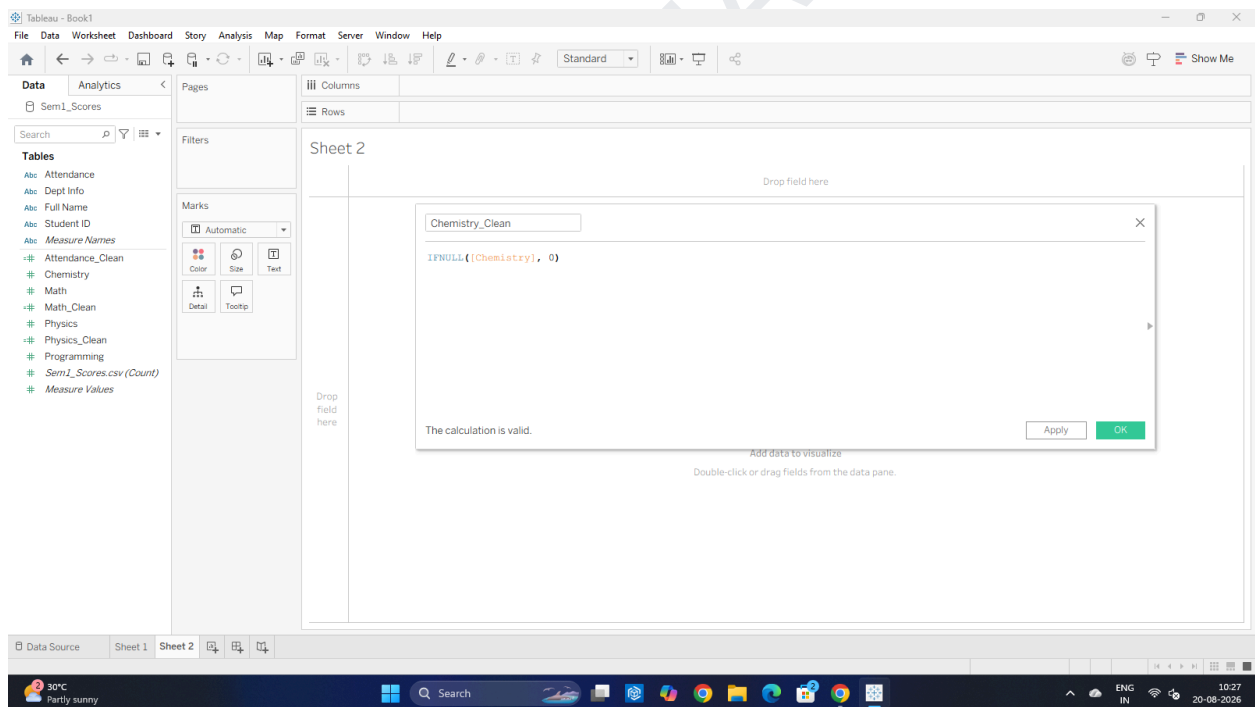
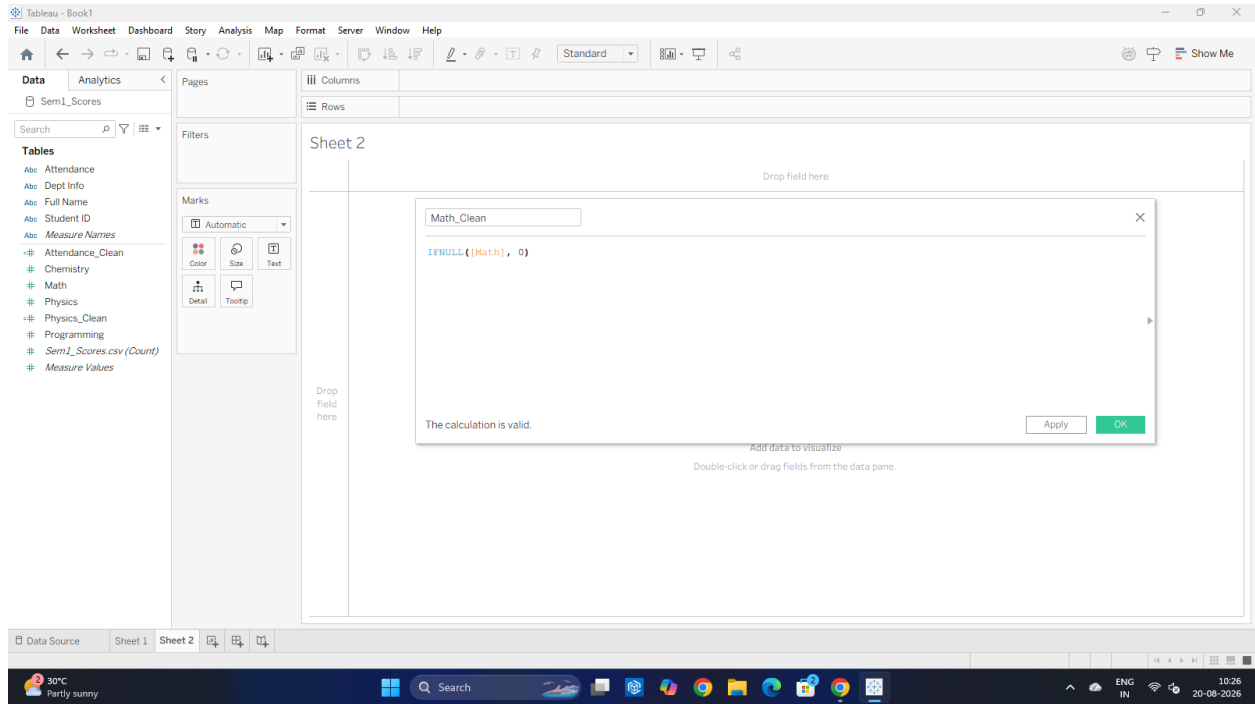
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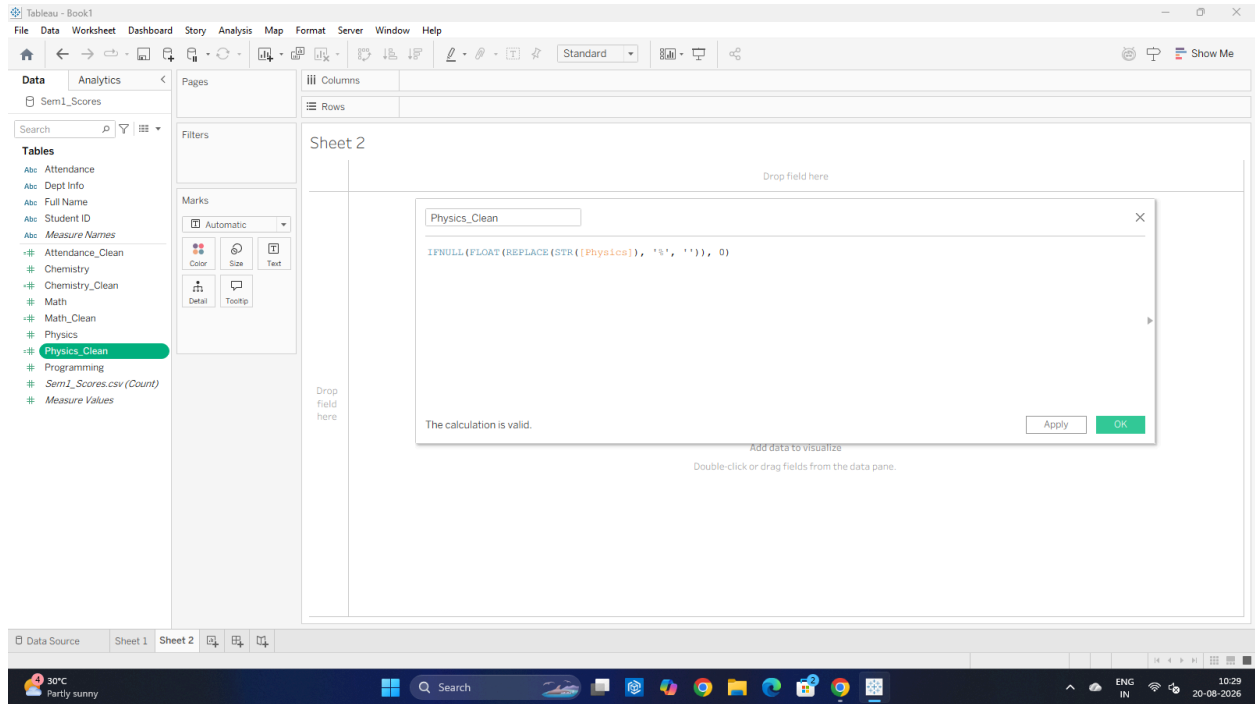
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b) Convert wide data into tall format

Tableau Desktop interface showing the 'Sem1_Scores' data source. The data is displayed in a wide format with 12 fields and 10 rows. The fields include Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The data is organized into a table with columns for each subject and a final column for Attendance.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

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The first screenshot shows the initial state of Tableau Desktop. The 'Sem1_Scores.csv' file is loaded as a data source. A context menu is open over the 'Fields' shelf, showing options like 'Rename', 'Reset Names', 'Copy Values', 'Hide', 'Create Calculated Field...', and 'Pivot'. The 'Pivot' option is selected, and the 'Fields' shelf is being updated to show a pivot table structure.

The second screenshot shows the final pivot table view. The 'Fields' shelf now contains the following fields:

- Row Labels: Attendance
- Columns: Subject, Score, Physics_Clean, Attendance_Clean, Math_Clean, Chemistry_Clean

The resulting pivot table is displayed below the shelves. It shows the percentage of students who attended, scored, and were clean in each subject.

Attendance	Subject	Score	Physics_Clean	Attendance_Clean	Math_Clean	Chemistry_Clean
85 %	Full Name	Aarav Sharma	null	85.0000	null	null
85 %	Math	85	null	85.0000	null	null
85 %	Physics	90%	null	85.0000	null	null
85 %	Programming	95	null	85.0000	null	null
85 %	Student ID	S101	null	85.0000	null	null
90 %	Chemistry	80	null	90.0000	null	null
90 %	Dept Info	ECE-02	null	90.0000	null	null

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The first screenshot shows the Tableau interface with 'Sem1_Scores' as the data source. A context menu is open over the 'Score' field, showing options like 'Number (decimal)', 'Date & Time', 'String', etc. The data preview table is visible below.

Score	Full Name	Student ID	Dept Info
85 %	Aarav Sharma	S101	ECE-02
85 %	Math		
85 %	Physics		
85 %	Programming		
85 %	Student ID	S101	
90 %	Chemistry		
90 %	Dept Info	ECE-02	

The second screenshot shows the 'Marks' card in the Tableau view. A calculation dialog box is open for 'Clean_Score' with the formula: `IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)`. The dialog confirms the calculation is valid.

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c) Perform union of multiple files

The top screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source selected. A 'Union' dialog box is open, showing the 'Specific (manual)' tab. The 'Connection: Sem1_Scores' is selected, and the 'Tables in union: 2' are listed: 'Sem2_Scores.Csv' and 'Sem1_Scores.csv'. The 'OK' button is highlighted.

The bottom screenshot shows the resulting 'Union' table with 9 fields and 20 rows. The table displays student data from multiple semesters, including Student ID, Full Name, Department, and scores in various subjects.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

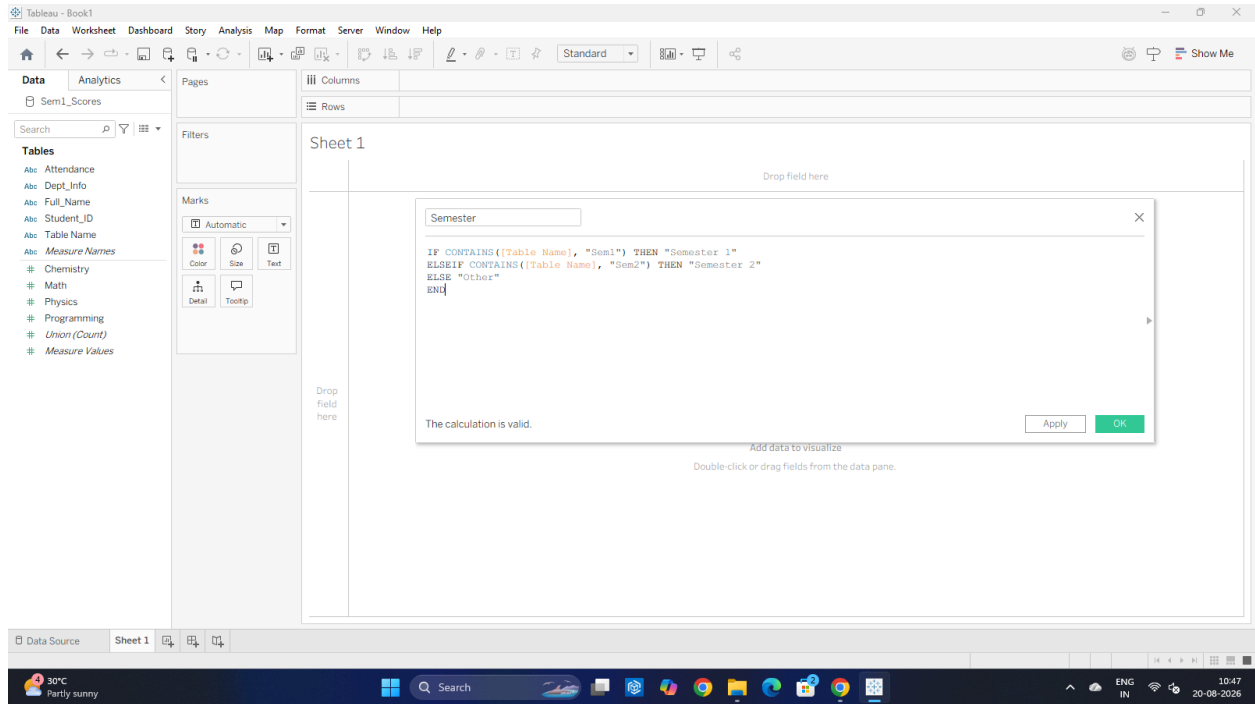
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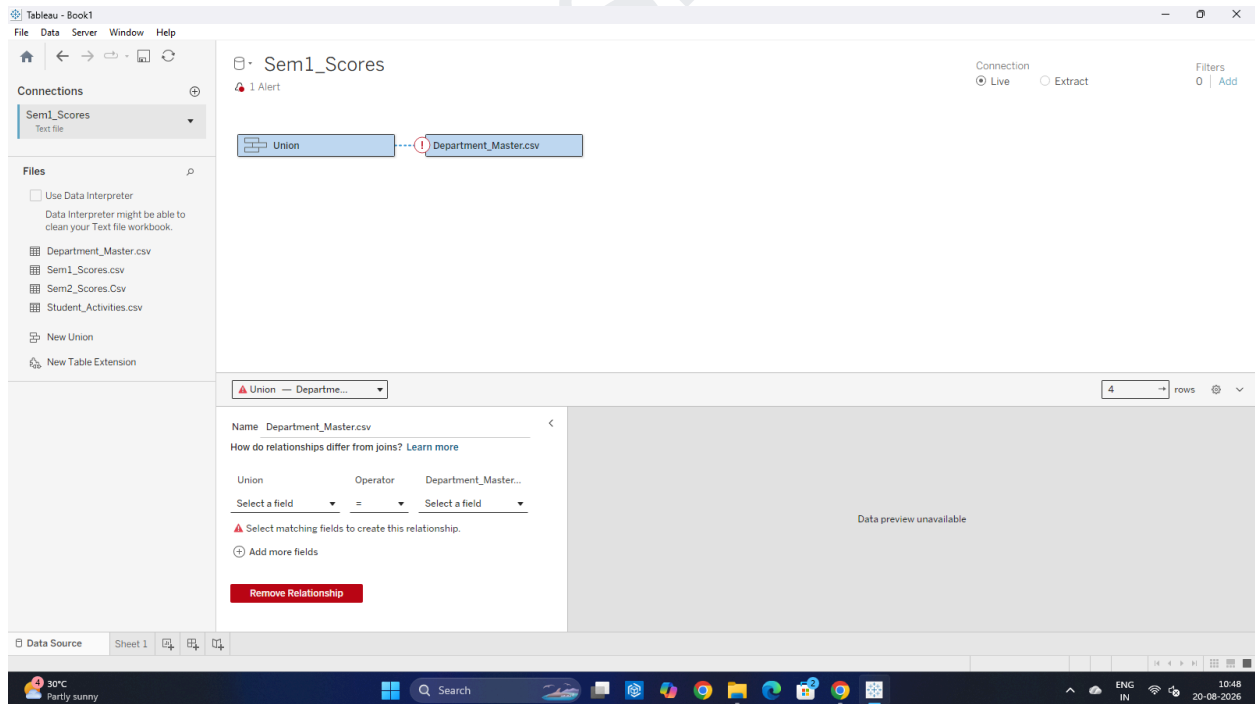
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d) Execute cross-database joins



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The first screenshot shows the Tableau interface with the 'Sem1_Scores' worksheet. The 'Connections' pane on the left lists 'Sem1_Scores' as a text file. The 'Files' pane shows several CSV files, including 'Department_Master.csv'. The main view displays a data table with columns: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, and Attendance. The table contains 7 rows of data.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

The second screenshot shows the same Tableau interface, but with a different data source connection. The 'Connections' pane now shows 'Sem1_Scores' as a text file. The 'Files' pane shows 'Student_Activities.csv'. The main view displays a data table with columns: Student ID (Student Activ), Club Membership, Projects Completed, and Certifications. The table contains 7 rows of data.

Student ID (Student Activ)	Club Membership	Projects Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2
S107	Robotics Club	2	1

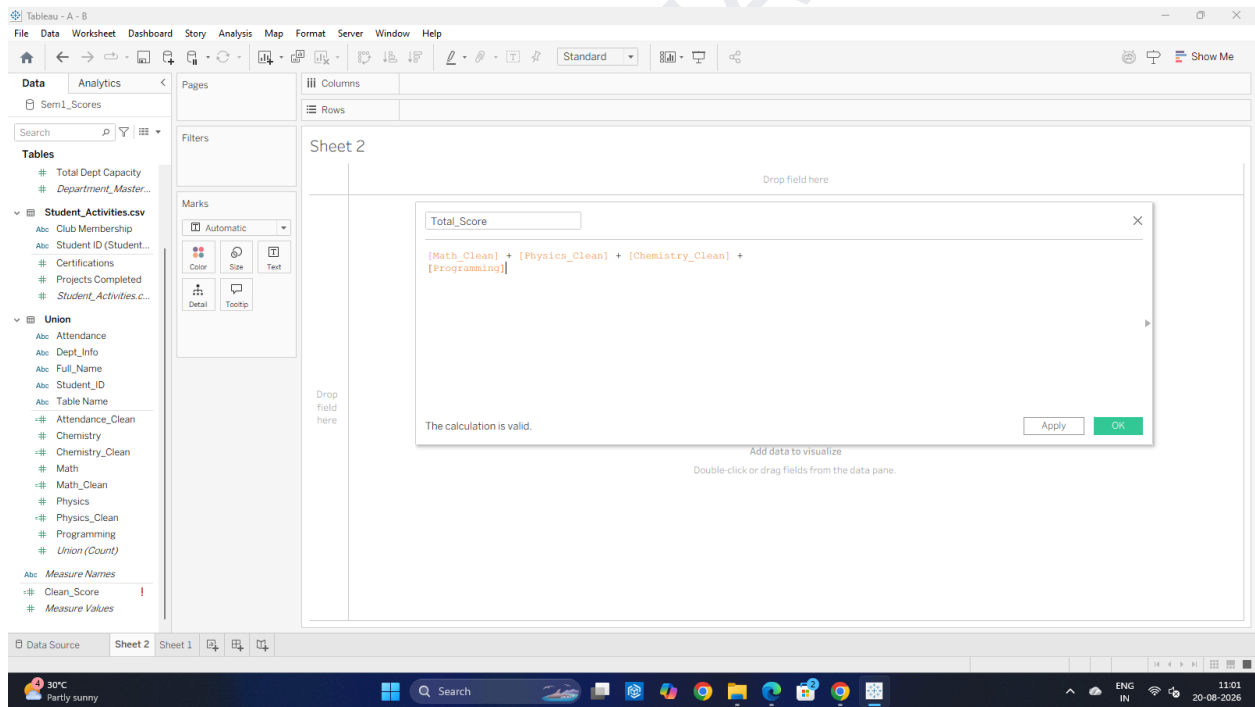
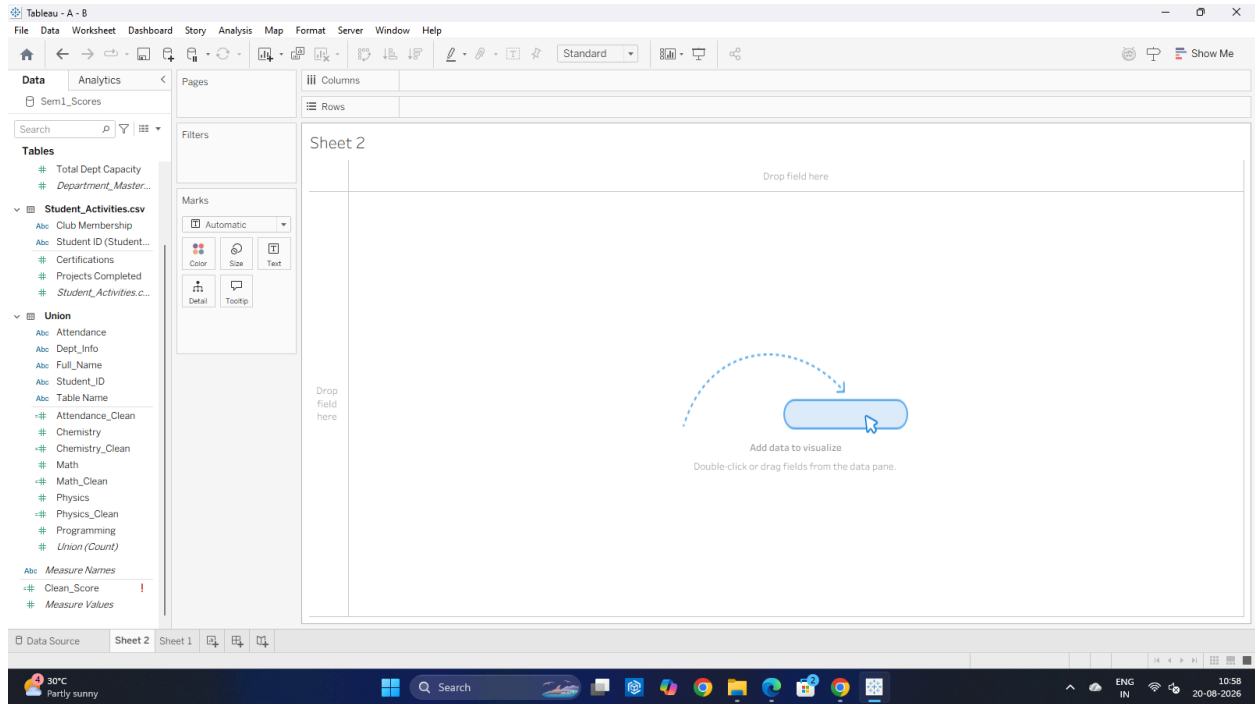
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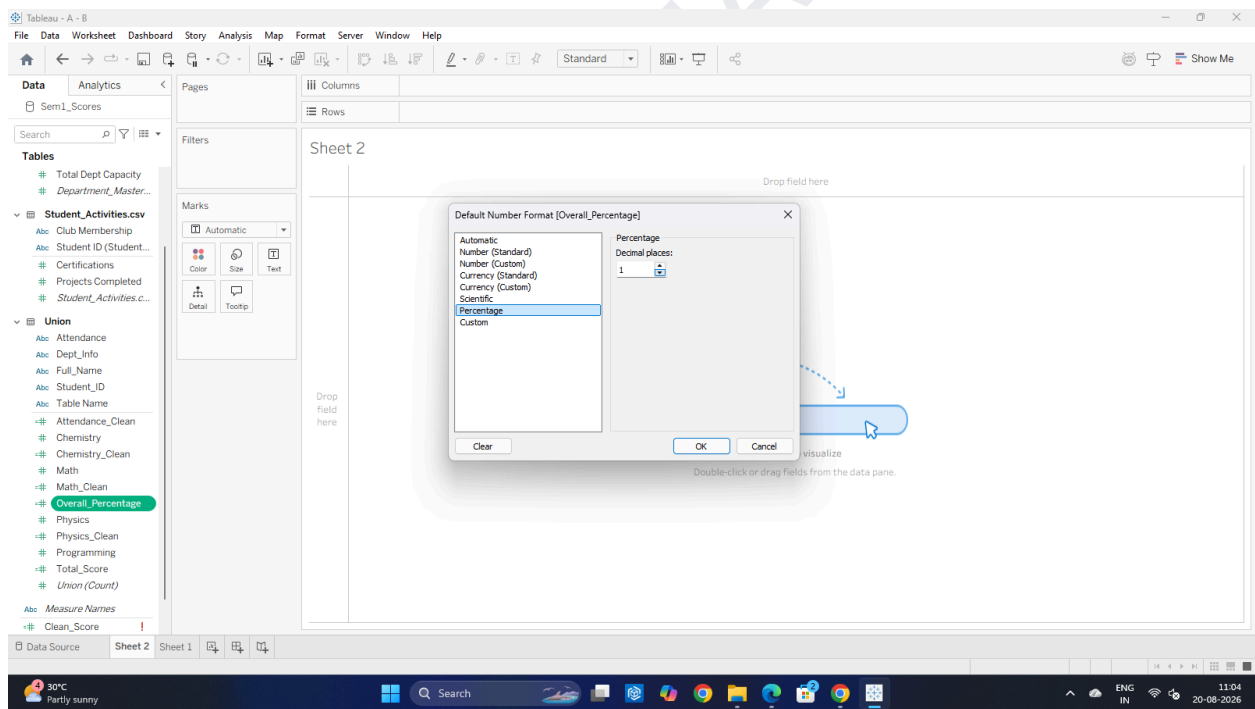
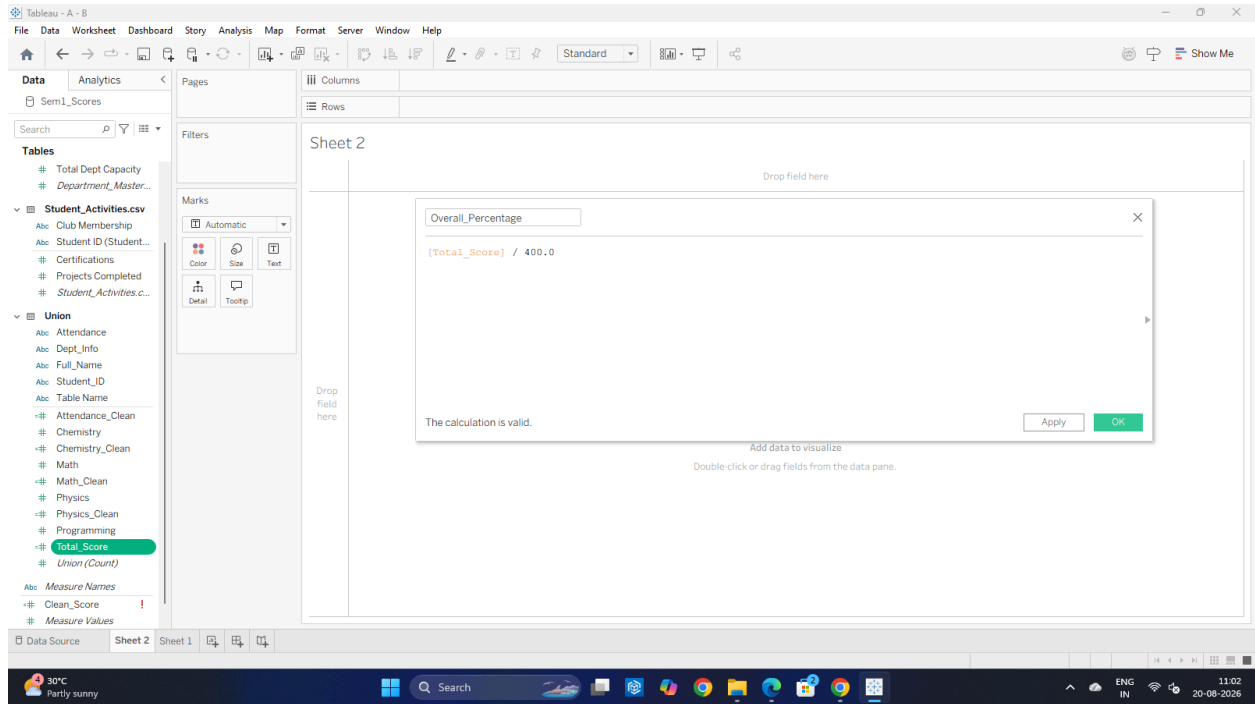
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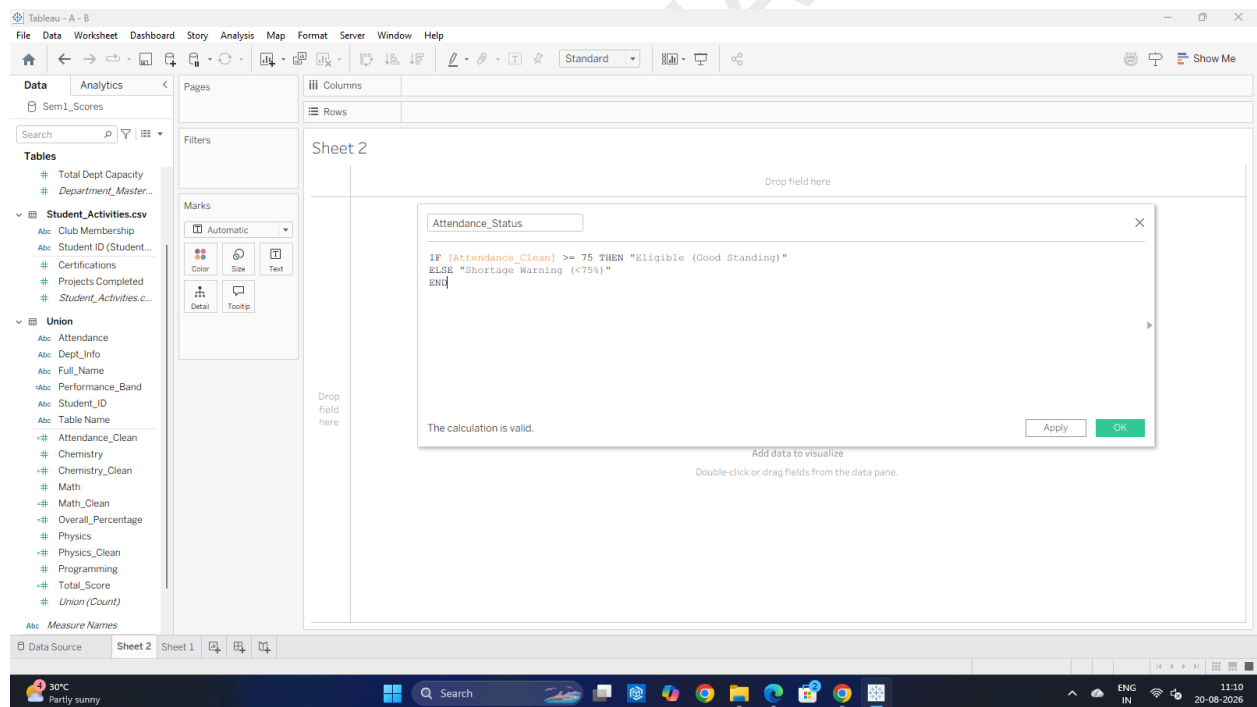
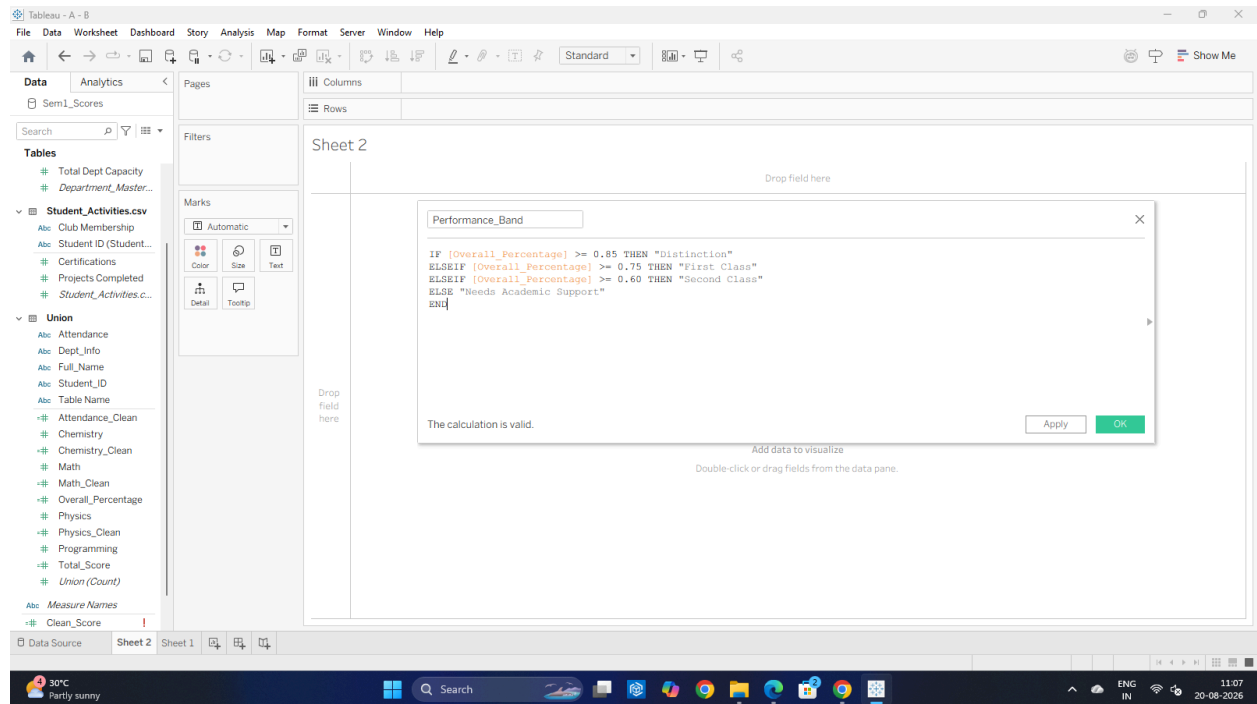
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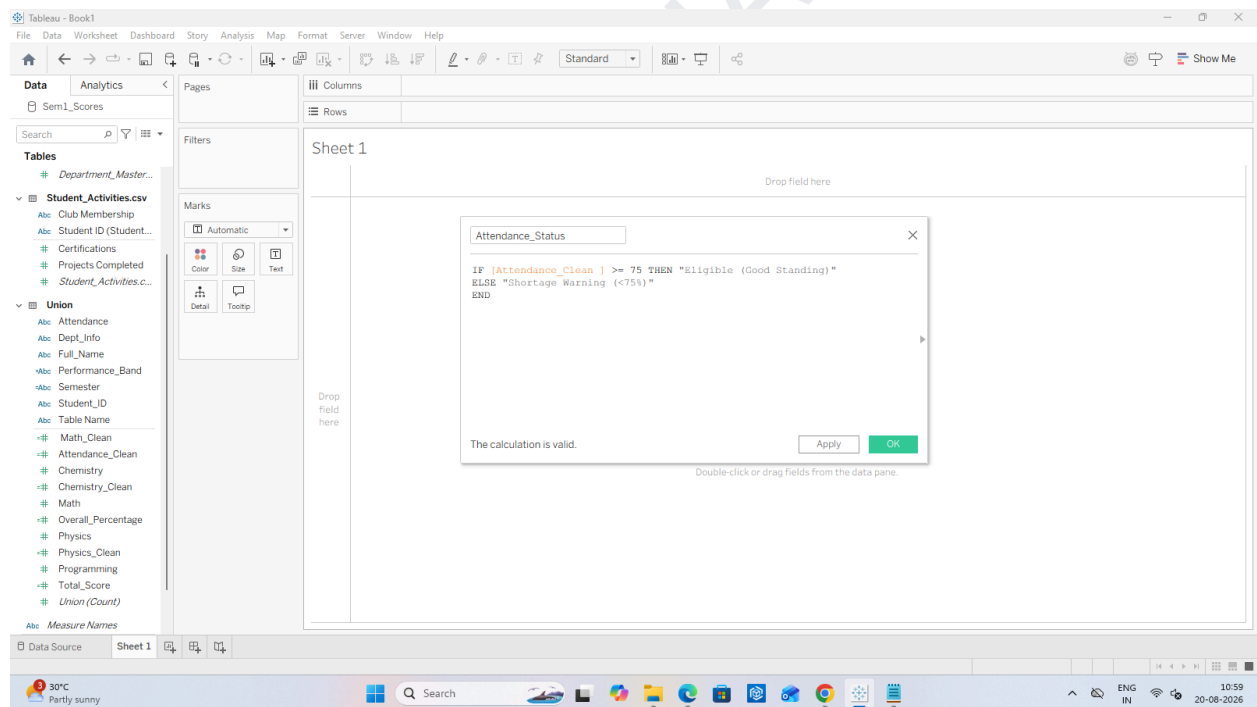
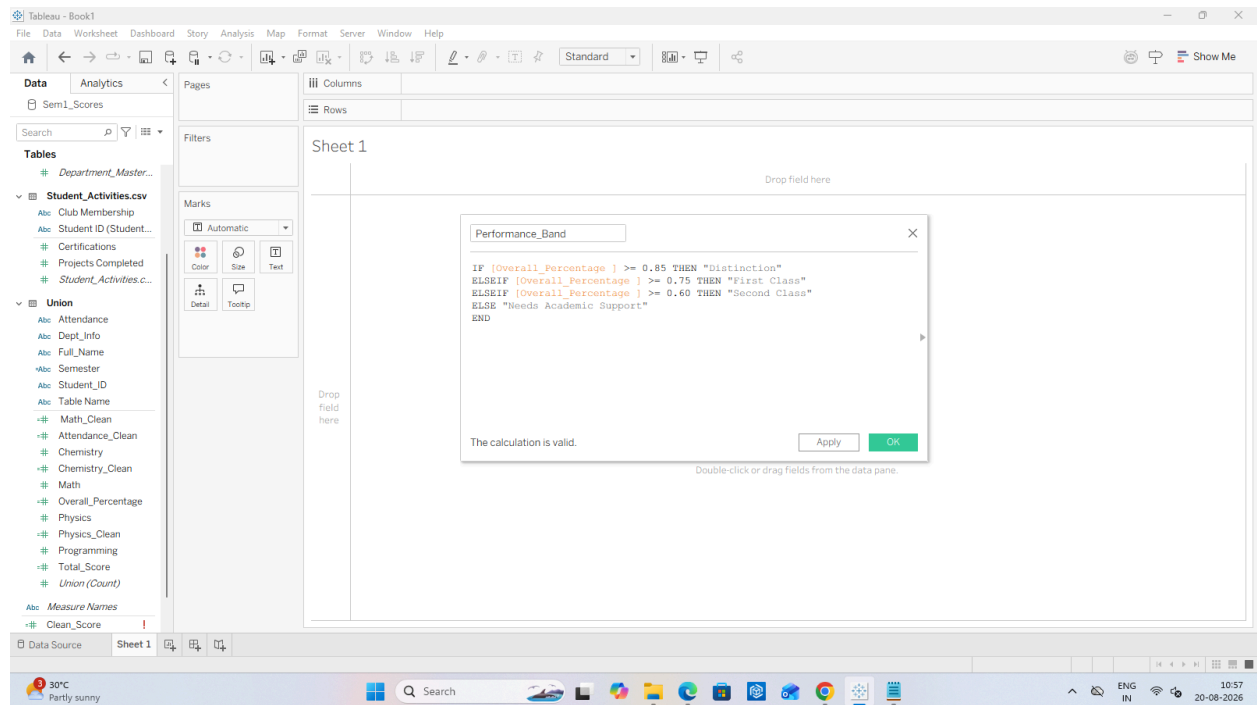
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e) Restructure data for better visualization



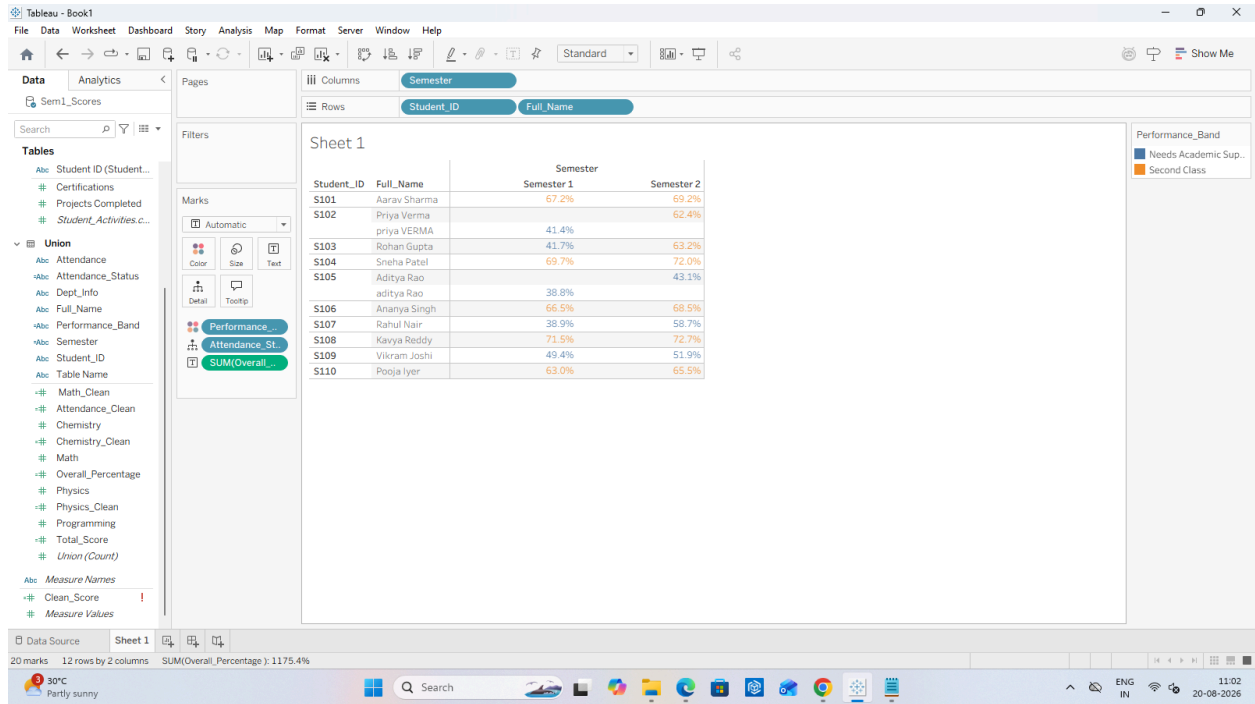
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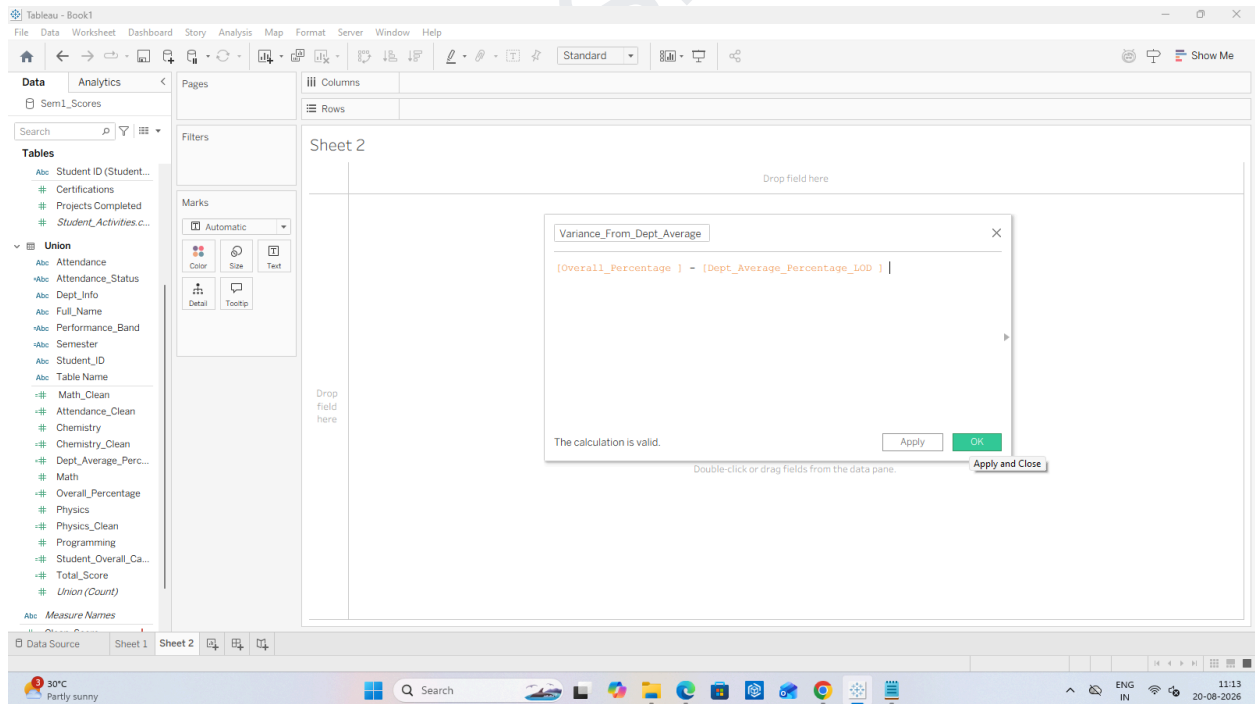
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f) Handle different levels of detail in datasets



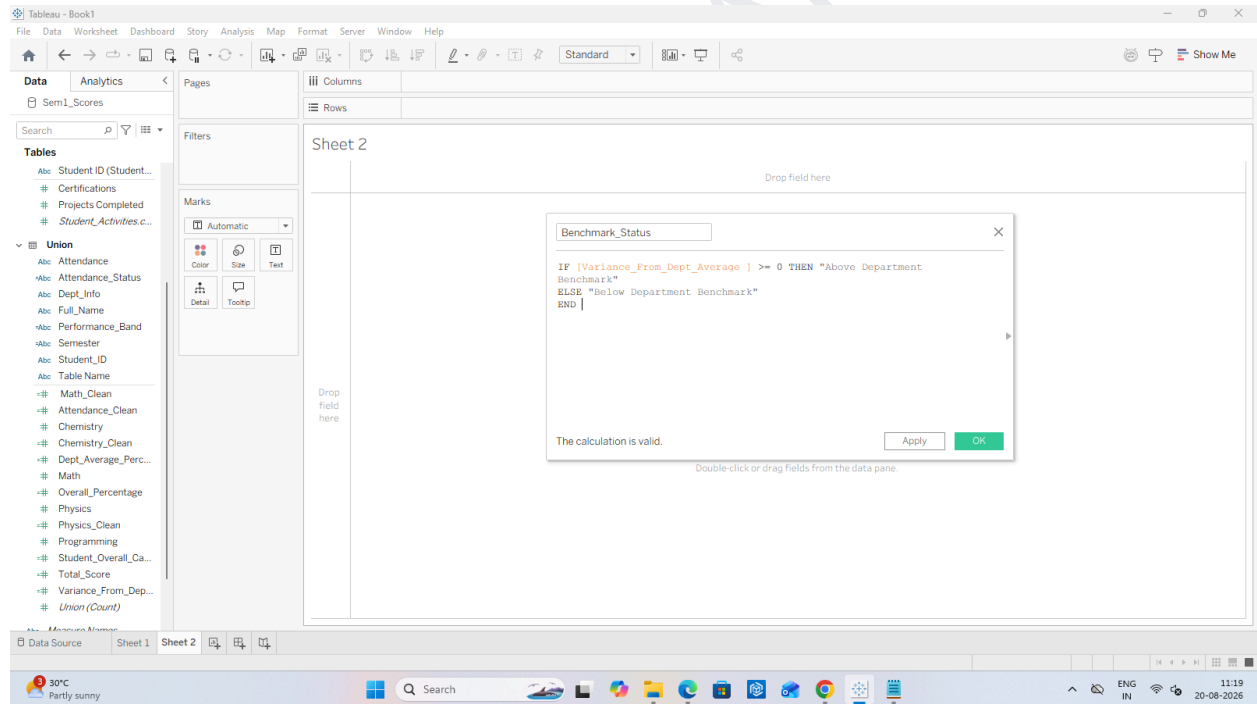
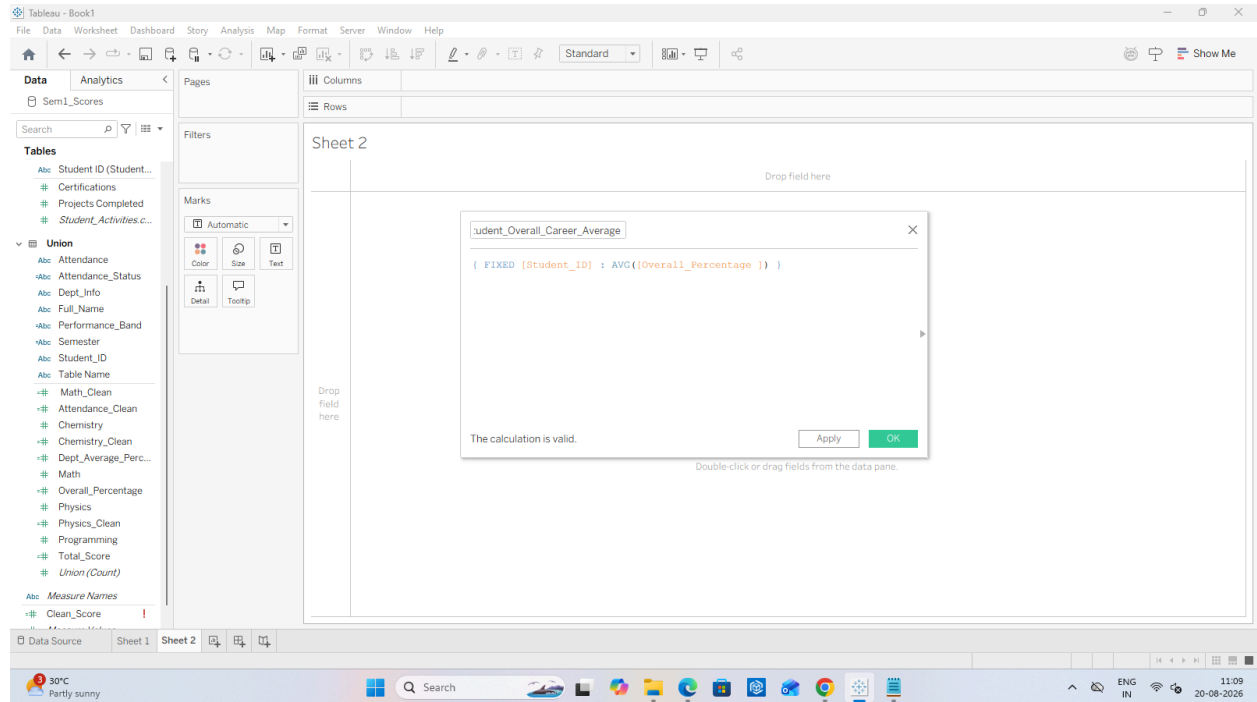
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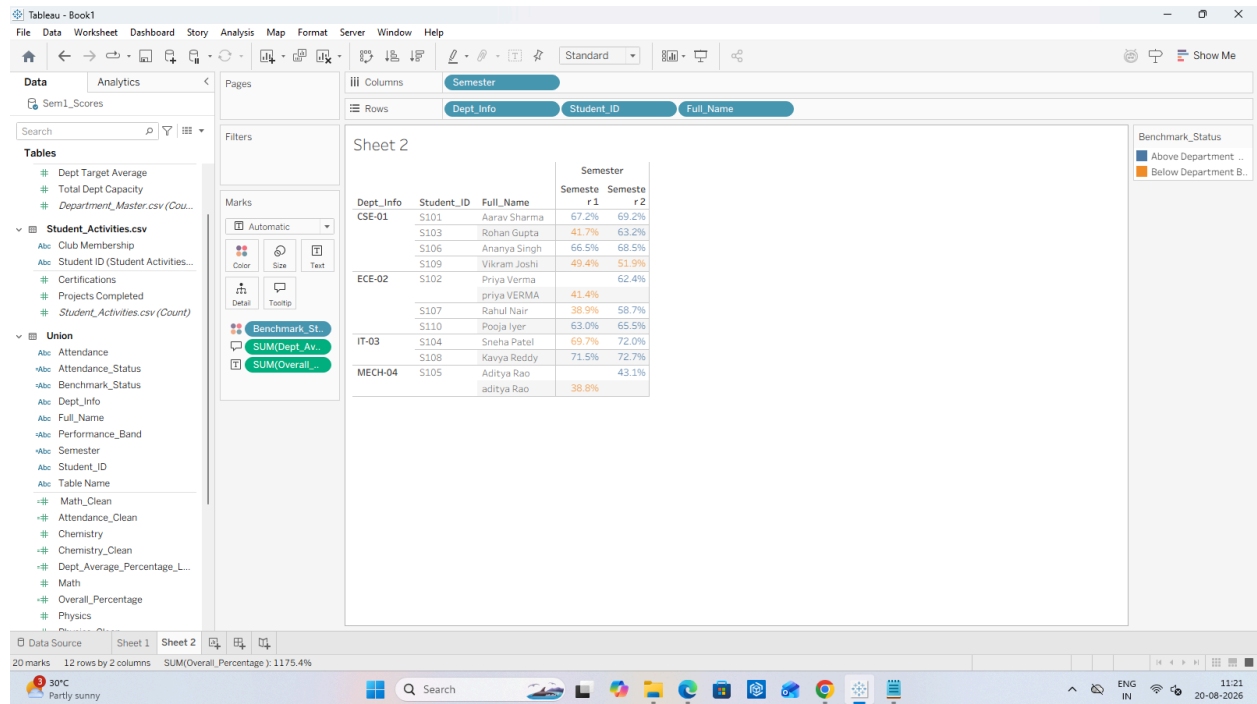
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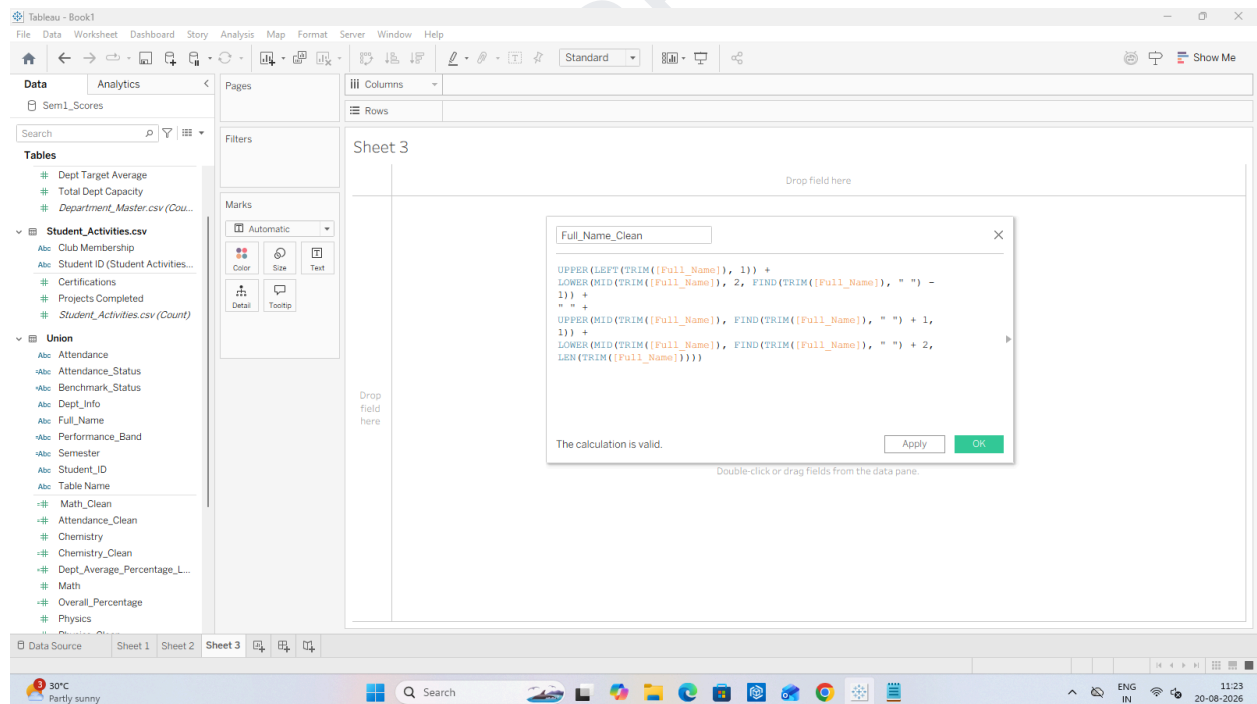
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g) Apply advanced data cleaning techniques



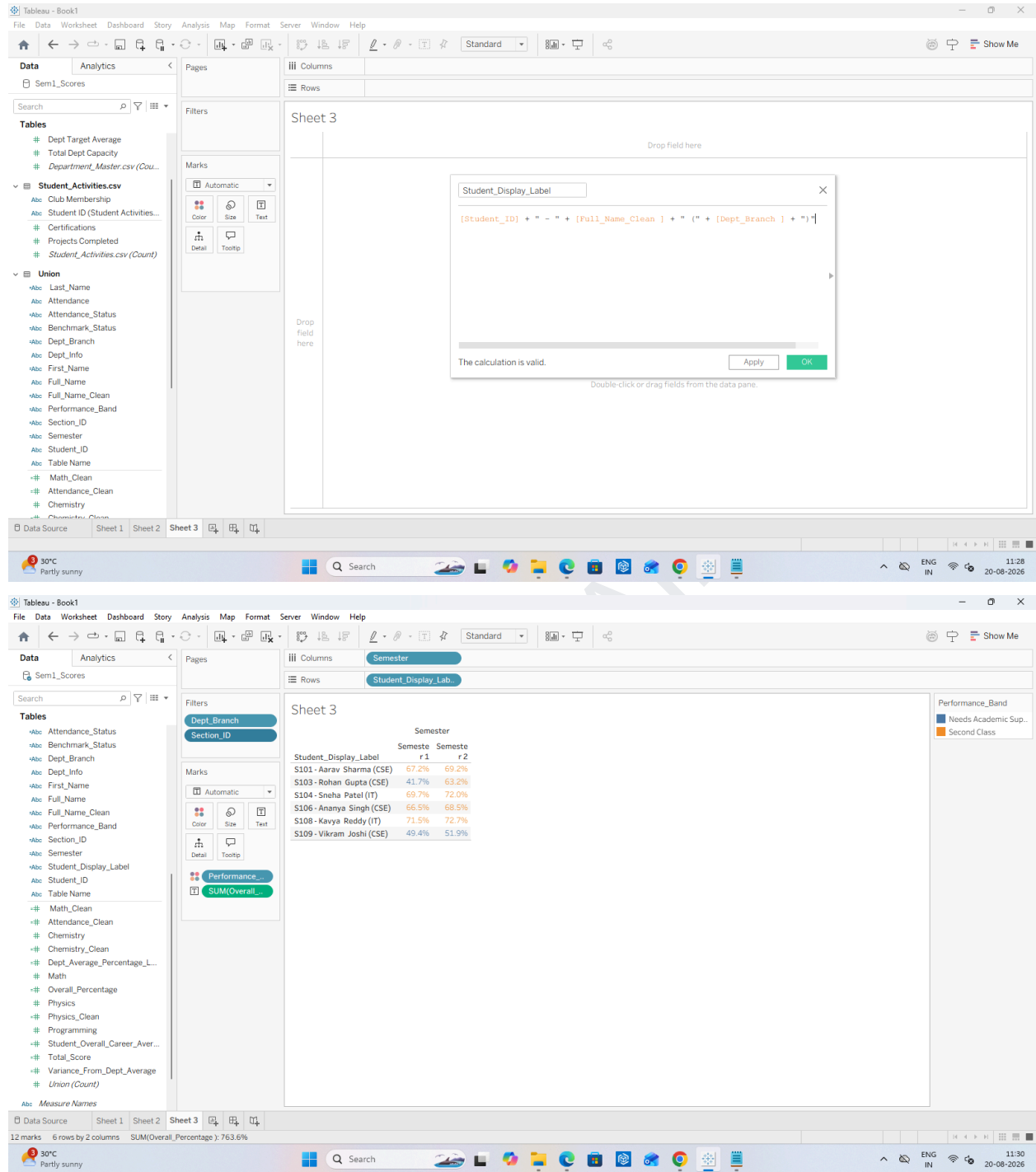
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